Class13FebSTAT.R

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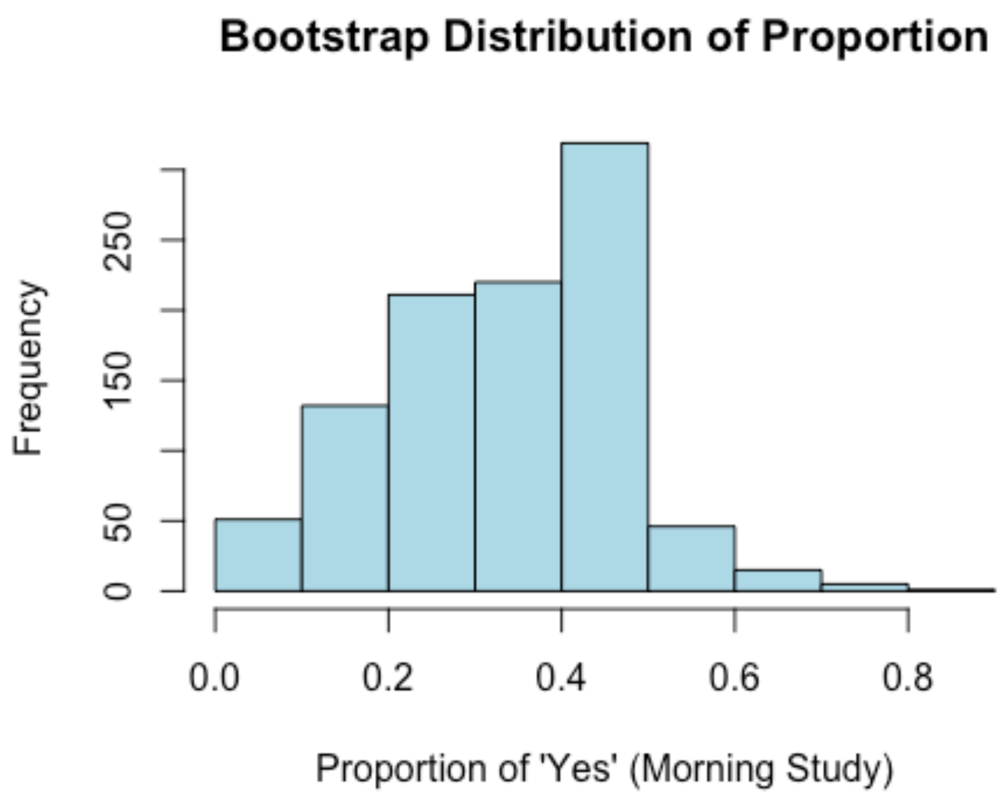
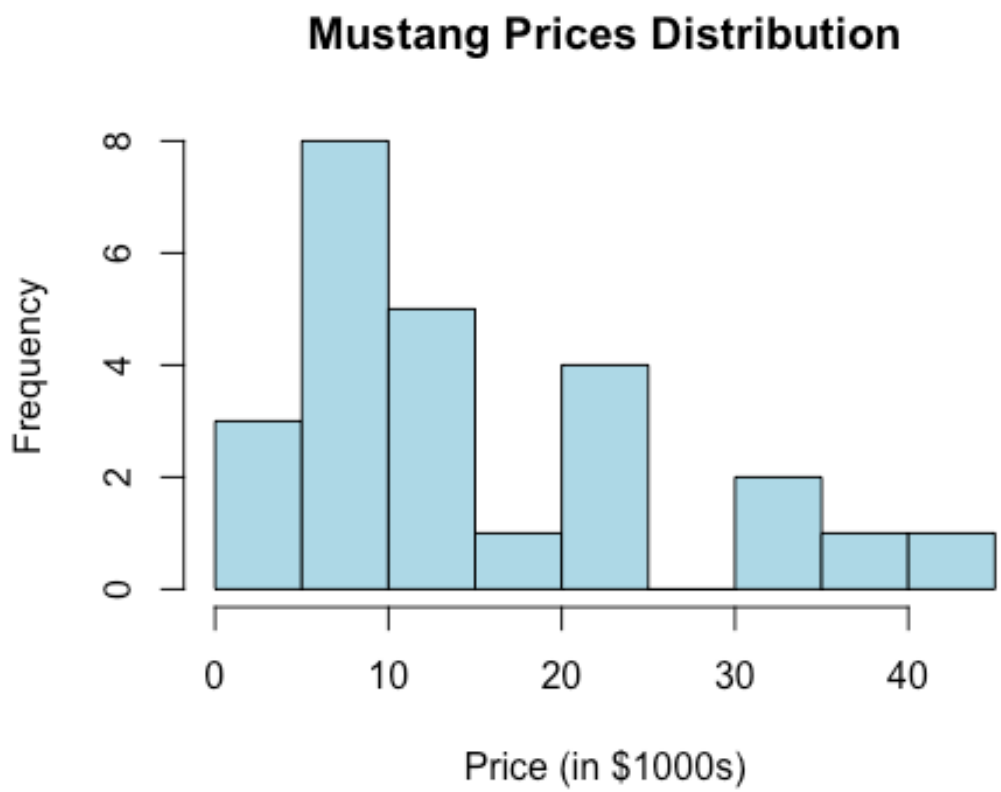
```
library(Lock5Data)
library(tidyverse)

## ── Attaching core tidyverse packages ──────────────────────── tidyverse
2.0.0 ──
## ✔ dplyr     1.1.4     ✔ readr     2.1.5
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ ggplot2   3.5.1     ✔ tibble    3.2.1
## ✔ lubridate 1.9.4     ✔ tidyr     1.3.1
## ✔ purrr     1.0.2
## ── Conflicts ──────────────────────────────────────────
tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force
all conflicts to become errors

# Question 1
# Part A Sample data: responses of students to whether they prefer studying
in the morning
responses <- c("yes", "yes", "yes", "yes", "no", "no", "no", "no", "no",
"no", "no", "no")
responses_numeric <- ifelse(responses == "yes", 1, 0)
prop_yes <- mean(responses_numeric)
prop_yes

## [1] 0.3333333

#Part B  Set seed
set.seed(123)
bootstrap_sample <- sample(responses_numeric, size=length(responses_numeric),
replace=TRUE)
prop_bootstrap <- mean(bootstrap_sample)
prop_bootstrap

## [1] 0.3333333

#Part C Generate 1000 bootstrap samples
n_bootstrap <- 1000
bootstrap_proportions <- replicate(n_bootstrap, {
  bootstrap_sample <- sample(responses_numeric,
size=length(responses_numeric), replace=TRUE)
  mean(bootstrap_sample)
```

```r
})
head(bootstrap_proportions)

## [1] 0.16666667 0.25000000 0.08333333 0.16666667 0.41666667 0.33333333

#Part D Visualize bootstrap distribution with histogram
hist(bootstrap_proportions, main="Bootstrap Distribution of Proportion",
xlab="Proportion of 'Yes' (Morning Study)", col="lightblue", border="black")
```



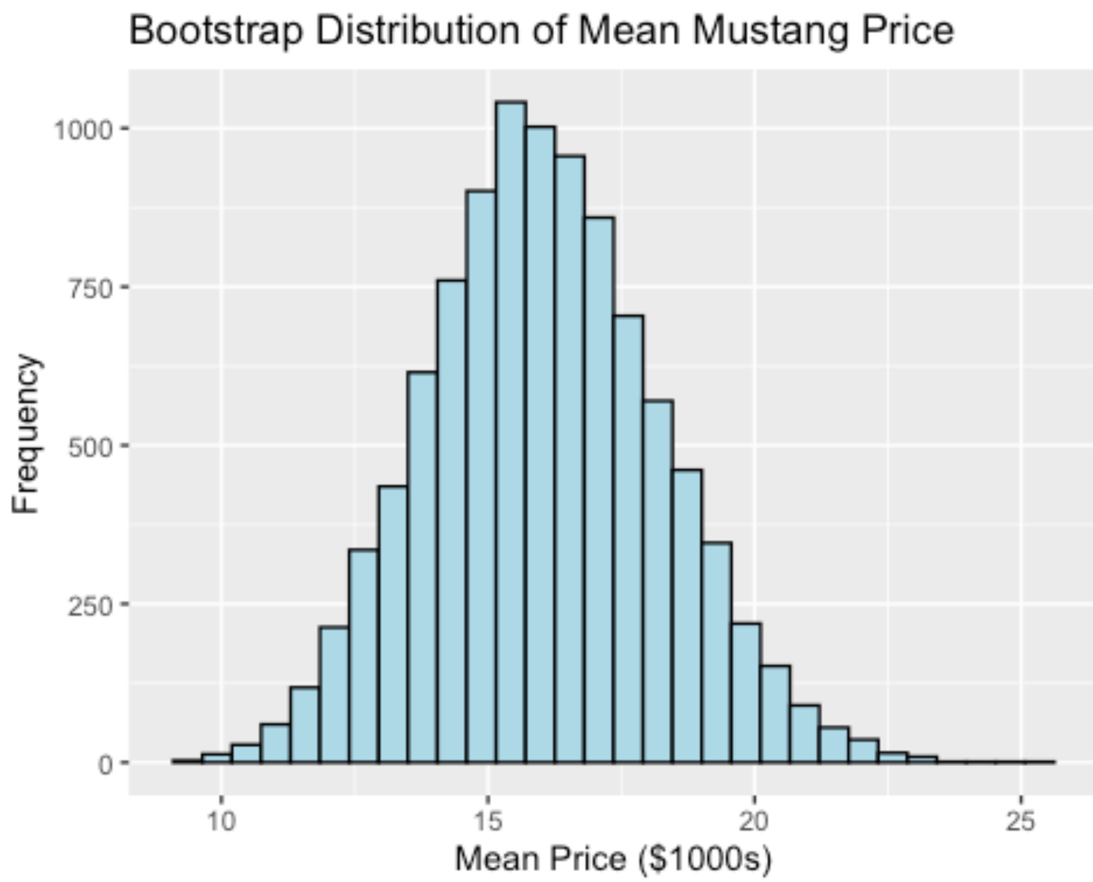

```r
mean_bootstrap <- mean(bootstrap_proportions)
mean_bootstrap

## [1] 0.3363333

#Part E Calculate 90% confidence interval
ci_lower <- quantile(bootstrap_proportions, 0.05)
ci_upper <- quantile(bootstrap_proportions, 0.95)
c(ci_lower, ci_upper)

##         5%        95%
## 0.08333333 0.58333333

# Question 2. (18 pts) The dataset "MustangPrice"
# Question A Mean and Standard Deviation
```

```
data("MustangPrice")
MustangPrice$Price

##  [1] 32.0 45.0 11.9 24.8 22.0 10.0  5.0  9.0 23.0 37.9 32.5  3.0  9.0 13.0
14.9
## [16]  7.0 16.0 21.0  7.0  8.2  9.7  8.0 11.8 12.9  4.9

n <- length(MustangPrice$Price)
mean_price <- mean(MustangPrice$Price)
sd_price <- sd(MustangPrice$Price)
print(mean_price)

## [1] 15.98

print(sd_price)

## [1] 11.11362

hist(MustangPrice$Price, main = "Mustang Prices Distribution", xlab = "Price
(in $1000s)", col = "lightblue")
```




```
# Question B thru D Bootstrap and Visualation
set.seed(123)
B <- 10000
bootstrap_means <- replicate(B, mean(sample(MustangPrice$Price, size = n,
```

```r
replace = TRUE)))
ggplot(data.frame(bootstrap_means), aes(x = bootstrap_means)) +
  geom_histogram(color = "black", fill = "lightblue", bins = 30) +
  labs(title = "Bootstrap Distribution of Mean Mustang Price", x = "Mean
Price ($1000s)", y = "Frequency")
```




```r
# Question E thru F  95% interval and Standard Error
CI_95_percentile <- quantile(bootstrap_means, c(0.025, 0.975))
SE_bootstrap <- sd(bootstrap_means)
CI_95_SE <- mean_price + c(-1.96, 1.96) * SE_bootstrap

# See results
cat("Sample Mean:", mean_price, "\n")

## Sample Mean: 15.98

cat("Sample Standard Deviation:", sd_price, "\n")

## Sample Standard Deviation: 11.11362

cat("95% Confidence Interval (Percentile Method):", CI_95_percentile, "\n")

## 95% Confidence Interval (Percentile Method): 11.936 20.5242
```

```r
cat("95% Confidence Interval (SE Method):", CI_95_SE, "\n")

## 95% Confidence Interval (SE Method): 11.70009 20.25991
```